

Name: _____

R. Hammack

Score: _____

Directions: This is a take-home test. It is due at the beginning of class on Wednesday, September 6. Please answer all questions in the space provided. Consider working the problems on scratch paper, then rewriting them neatly on the test. Additional copies of this test can be downloaded from my web page if needed.

For this test, you may discuss the problems among yourselves and share ideas, but the work you turn in must be your own (not copied). At the end of your solution to each problem, please list who (if anyone) you talked to about that problem, plus any additional information you want me to know (i.e. that you *gave* more help than you *received*, or vice versa, etc.).

- Use Gaussian elimination or Gauss-Jordan elimination to solve the problems.
- Please clearly indicate all of your row operations (e.g. $R_2 + 3R_4 \rightarrow R_3$, etc).
- Put your final matrix in row-echelon or reduced row-echelon form.
- If a system has more than one solution, state the solution set in **parametric form**.
- Indicate your solution clearly by putting it in a box.
- Constants that are not integers should be expressed as fractions.
- You may consult your text and notes, but **no** other source.
- In order to get full credit, you must show all of your work.

1. (8 points) Solve the system
$$\begin{cases} 2x + 4y = 7 \\ 6x - 2y = 0 \end{cases}$$

2. (8 points) Solve the system
$$\begin{cases} 2x + 4y = 7 \\ 6x + 12y = 0 \end{cases}$$

3. (8 points) Solve the system
$$\begin{cases} 2x + 4y = 7 \\ 6x + 12y = 21 \end{cases}$$

4. (12 points) Solve the system
$$\begin{cases} 3x - 2y = 0 \\ x - 2y = -4 \\ x - y = -1 \\ 3x + 2y = 12 \end{cases}$$

5. (12 points) Solve the system
$$\begin{cases} 5x_1 - 10x_2 - 5x_3 + 15x_4 = 25 \\ -3x_1 + 6x_2 + 2x_3 + x_4 = 5 \end{cases}$$

6. (12 points) Solve the system
$$\begin{cases} 2w & - & 2x & & + & 2z & = & 4 \\ w & + & x & - & y & - & z & = & 2 \\ -w & + & x & - & y & - & z & = & 2 \\ w & - & x & - & y & + & z & = & 2 \end{cases}$$

7. (12 points) Solve the system
$$\begin{cases} -3x & + & 6y & - & 9z & = & 12 \\ x & - & 2y & + & 3z & = & -4 \\ -\frac{1}{3}x & + & \frac{2}{3}y & - & z & = & \frac{4}{3} \end{cases}$$

8. (12 points) Solve the system
$$\begin{cases} x & + & 3y & + & 2z & = & 1 \\ 2x & + & 9y & - & z & = & 8 \\ 3x & - & 6y & + & 4z & = & 0 \end{cases}$$

9. (16 points) Solve the system $\left\{ \begin{array}{cccccccc} x_1 & + & 3x_2 & - & 2x_3 & & + & 2x_5 & & = & 0 \\ 2x_1 & + & 6x_2 & - & 5x_3 & - & 2x_4 & + & 4x_5 & - & 3x_6 & = & -1 \\ & & & & 5x_3 & + & 10x_4 & & & + & 15x_6 & = & 5 \\ 2x_1 & + & 6x_2 & & & + & 8x_4 & + & 4x_5 & + & 18x_6 & = & 6 \end{array} \right.$